

2008

**HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Physics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Draw diagrams using pencil
- Board-approved calculators may be used
- A data sheet, formulae sheets and Periodic Table are provided at the back of this paper
- Write your Centre Number and Student Number at the top of pages 9, 11, 13, 15, 17 and 19

Total marks – 100

Section I Pages 2–22

75 marks

This section has two parts, Part A and Part B

Part A – 15 marks

- Attempt Questions 1–15
- Allow about 30 minutes for this part

Part B – 60 marks

- Attempt Questions 16–27
- Allow about 1 hour and 45 minutes for this part

Section II Pages 23–34

25 marks

- Attempt ONE question from Questions 28–32
- Allow about 45 minutes for this section

Section I

75 marks

Part A – 15 marks

Attempt Questions 1–15

Allow about 30 minutes for this part

Use the multiple-choice answer sheet for Questions 1–15.

- 1 An object on Earth has a weight of 490 N and experiences an acceleration due to gravity of 9.8 m s^{-2} . On Mars, this object would experience an acceleration due to gravity of 3.7 m s^{-2} .

On Mars, what would be the weight of this object?

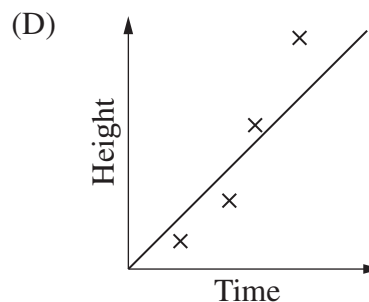
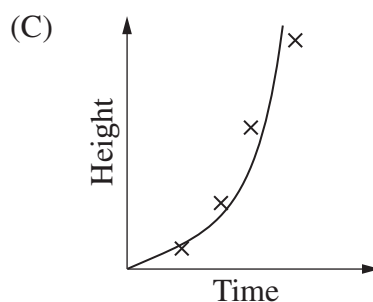
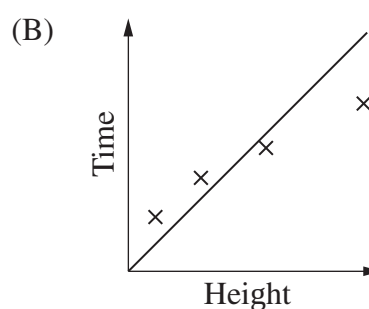
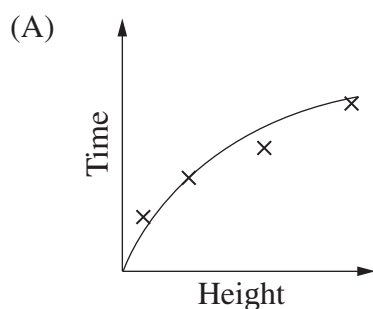
- (A) 490 N
- (B) $\frac{490}{9.8}$ N
- (C) $\frac{490}{9.8} \times 3.7$ N
- (D) $\frac{490}{3.7} \times 9.8$ N
- 2 Which of these statements best describes the forces acting on a satellite in orbit around Earth?
- (A) Although gravity has no effect, there is still an outward force.
- (B) The satellite is kept up by an outward force that balances the force due to gravity.
- (C) Gravity is the only force acting on the satellite and this results in an inward acceleration.
- (D) The effect of gravity is negligible, the satellite is kept in orbit by its momentum and the net force on it is zero.

- 3 An aeroplane is flying horizontally over level ground. It has an altitude of 490 m and a velocity of 100 m s^{-1} . As the aeroplane passes directly above a cross marked on the ground, an object is released from the aeroplane.

How far away from the cross will this object land?

- (A) 490 m
(B) 1000 m
(C) 10 000 m
(D) 49 000 m
- 4 An investigation was performed to determine the acceleration due to gravity. A ball was dropped from various heights and the time it took to reach the ground from each height was measured. The results were graphed with the independent variable on the horizontal axis.

Which graph best represents the relationship between the variables?

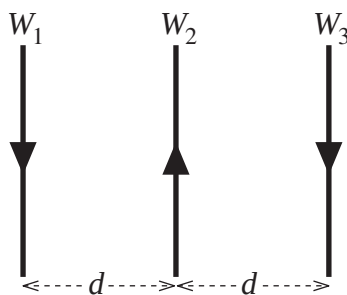


- 5 A spaceship is travelling away from Earth at $1.8 \times 10^8 \text{ m s}^{-1}$. The time interval between consecutive ticks of a clock on board the spaceship is 0.50 s. Each time the clock ticks, a radio pulse is transmitted back to Earth.

What is the time interval between consecutive radio pulses as measured on Earth?

- (A) 0.40 s
(B) 0.50 s
(C) 0.63 s
(D) 0.78 s

- 6 Three identical wires W_1 , W_2 and W_3 are positioned as shown. Each carries a current of the same magnitude in the direction indicated.

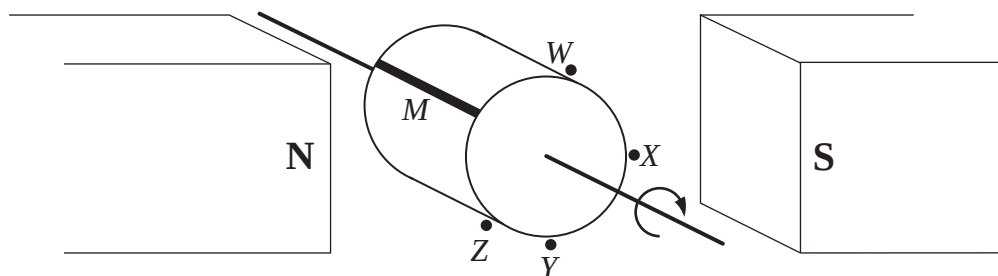


What is the magnitude and direction of the resultant force on W_2 ?

	<i>Magnitude</i>	<i>Direction</i>
(A)	Zero	None
(B)	Non zero	To the left
(C)	Non zero	To the right
(D)	Non zero	Out of the page

- 7 Which of the following is necessary for the operation of an AC induction motor?
- (A) A fixed magnetic field in the rotor
 - (B) A direct current supply to the rotor
 - (C) A changing magnetic field in the rotor
 - (D) Split rings conducting current to the rotor

- 8 A plastic cylinder with a metal strip, M , on its surface is rotated at constant speed about its axis, in a uniform magnetic field. During each rotation the strip, M , passes locations W , X , Y and Z shown below.



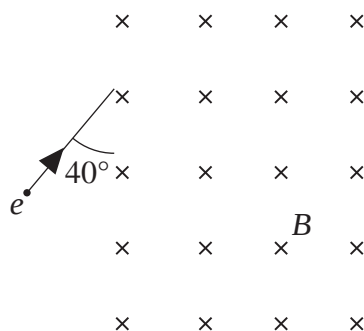
When is the potential difference across M greatest?

- (A) As M passes W .
 - (B) As M passes X .
 - (C) As M passes Y .
 - (D) As M passes Z .
- 9 Which statement best explains how induction cooktops heat food?
- (A) Eddy currents generated in the water in the food produce heat.
 - (B) Eddy currents generated in the base of the saucepan produce heat.
 - (C) Resistance in the glass of the cooktop produces heat.
 - (D) Resistance in the element beneath the glass cooktop produces heat.
- 10 The cathode ray tube and transistor circuits in a conventional television rely on transformers.

What transformation of the 240 V AC input voltage do these components require?

	<i>Cathode ray tube</i>	<i>Transistor circuits</i>
(A)	Step-up	Step-down
(B)	Step-down	Step-up
(C)	Step-up	Step-up
(D)	Step-down	Step-down

- 11 An electron, e , moving with a velocity of $8.0 \times 10^6 \text{ m s}^{-1}$ enters a uniform magnetic field, B , of strength $2.1 \times 10^{-2} \text{ T}$ as shown.



The electron experiences a force which causes it to move along a circular path.

What is the radius of the path followed by the electron?

- (A) $1.1 \times 10^{-3} \text{ m}$
 - (B) $1.4 \times 10^{-3} \text{ m}$
 - (C) $1.7 \times 10^{-3} \text{ m}$
 - (D) $2.2 \times 10^{-3} \text{ m}$
- 12 The debate as to whether cathode rays are charged particles or electromagnetic waves continued for many years.

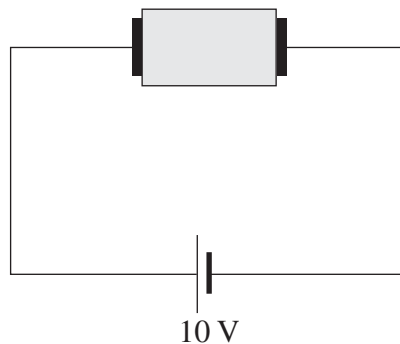
Which observation of cathode rays resolved this debate?

- (A) Cathode rays can turn a paddle wheel.
 - (B) An electric field can deflect cathode rays.
 - (C) Cathode rays can penetrate thin metal foil.
 - (D) Fluorescent screens glow when struck by cathode rays.
- 13 What is the energy of a photon of wavelength 580 nm ?
- (A) $3.43 \times 10^{-19} \text{ J}$
 - (B) $3.43 \times 10^{-28} \text{ J}$
 - (C) $3.85 \times 10^{-31} \text{ J}$
 - (D) $3.85 \times 10^{-40} \text{ J}$

- 14 When a magnet is released above a superconductor that has been cooled below its critical temperature, the magnet hovers above the superconductor. This is called the Meissner effect.

What is the best explanation for this?

- (A) The net force is zero due to electrostatic repulsion.
 - (B) The magnetic field freezes at very low temperature.
 - (C) The net force is zero due to repulsion between the Cooper pairs.
 - (D) The superconductor excludes magnetic fields at very low temperatures.
- 15 A block of silicon doped with boron is connected as shown in the diagram below.



What is the main way in which conduction occurs in the doped silicon block?

- (A) Valence band electrons move to the right.
- (B) Valence band electrons move to the left.
- (C) Conduction band electrons move to the right.
- (D) Conduction band electrons move to the left.

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Centre Number

Section I (continued)

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Part B – 60 marks

Attempt Questions 16–27

Allow about 1 hour and 45 minutes for this part

Student Number

Answer the questions in the spaces provided.

Show all relevant working in questions involving calculations.

Marks

Question 16 (3 marks)

Using a diagram and text, describe how an investigation can be performed to demonstrate the production and reception of radio waves.

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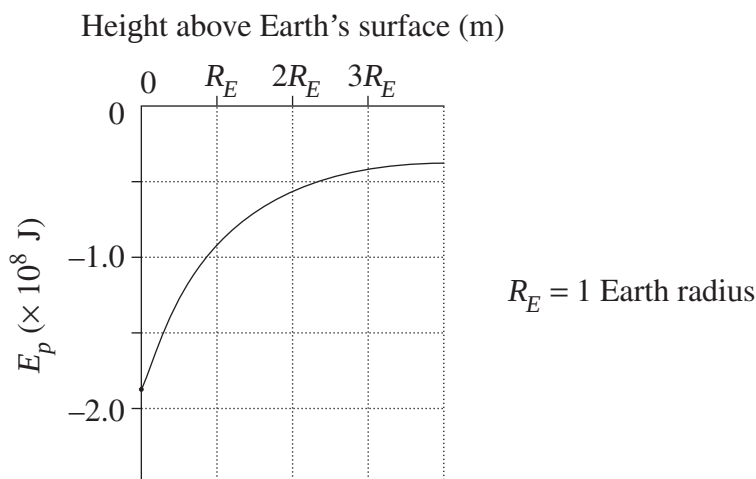
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Question 17 (5 marks)

The graph below represents the gravitational potential energy (E_p) of a mass as it is raised above Earth's surface.



- (a) From the graph, what is the gravitational potential energy of the mass when it is one Earth radius above Earth's surface? 1

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- (b) Use an equation to explain why the graph is a curve and not a straight line. 1

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- (c) Explain what happens to a rocket's chemical energy, kinetic energy and gravitational potential energy when it is being launched from the surface of Earth. 3

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Centre Number

Section I (continued)

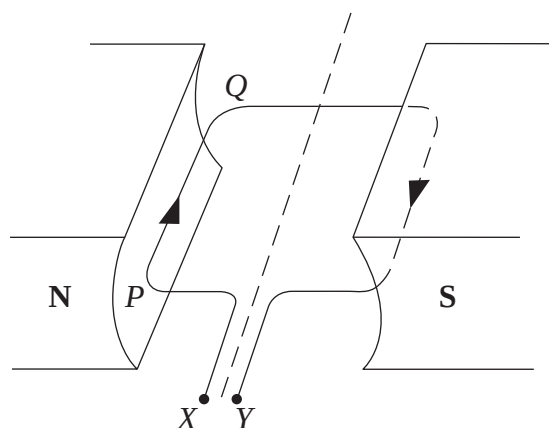
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Student Number

Marks

Question 18 (4 marks)

The diagram shows a coil in a magnetic field. The coil can rotate freely.



The coil is connected to a power supply and, at the instant shown, terminal X is positive.

- (a) In which direction will side PQ initially move?

1

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- (b) When the coil starts rotating, the potential difference experienced by the electrons in the wire is less than that supplied by the power supply.

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Describe the origin of this effect.

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Question 19 (8 marks)

- (a) Explain the changes in momentum when a satellite fires its propulsion system. 3

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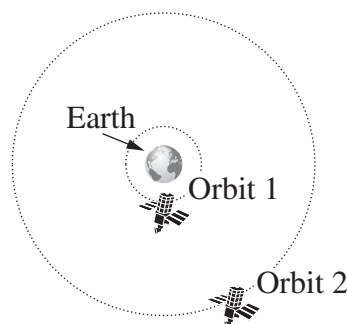
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- (b) A satellite is propelled from Orbit 1 to Orbit 2 as shown in the diagram.



- Orbit 2 has a radius of 27 000 km. What is the satellite's speed in this orbit? 3

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- (c) The radius of Orbit 2 is four times that of Orbit 1. What is the ratio of the new orbital period to the original period? 2

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Centre Number

Section I (continued)

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Student Number

Marks

Question 20 (4 marks)

Compare how electric current is conducted through samples of germanium at room temperature, mercury at room temperature and mercury at 3 K (T_c for mercury is 4.2 K).

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- space exploration

- large-scale electricity distribution systems.

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Centre Number

Section I (continued)

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Student Number

Marks

Question 22 (3 marks)

Explain why the development of transformers was necessary to enable the large-scale distribution of electrical power.

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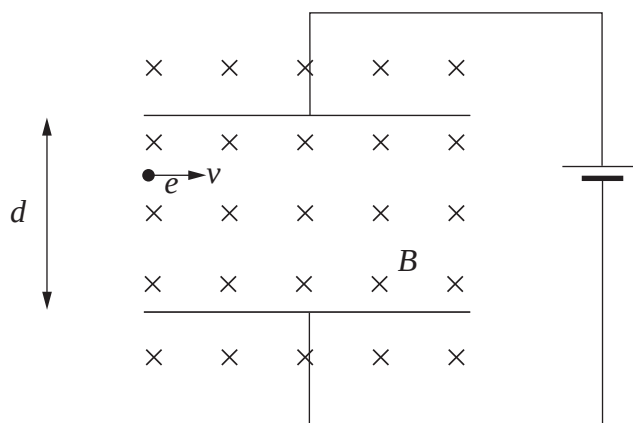
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Question 23 (7 marks)

Two parallel metal plates in a magnetic field are separated by a distance d , as shown. An electron enters the space between the plates.



- (a) On the diagram indicate with an arrow the direction of the force on the electron due to the magnetic field. 1

- (b) The strength of the magnetic field is $B = 0.001 \text{ T}$ and the electron's velocity is $v = 2 \times 10^6 \text{ m s}^{-1}$. Calculate the magnitude of the magnetic force on the electron. 2

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- (c) If $d = 10 \text{ mm}$, calculate the voltage required for the electron to continue on a straight path parallel to the plates. 2

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- (d) How was this experimental set-up used by Thomson to determine the charge/mass ratio of an electron? 2

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Section I (continued)

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Student Number

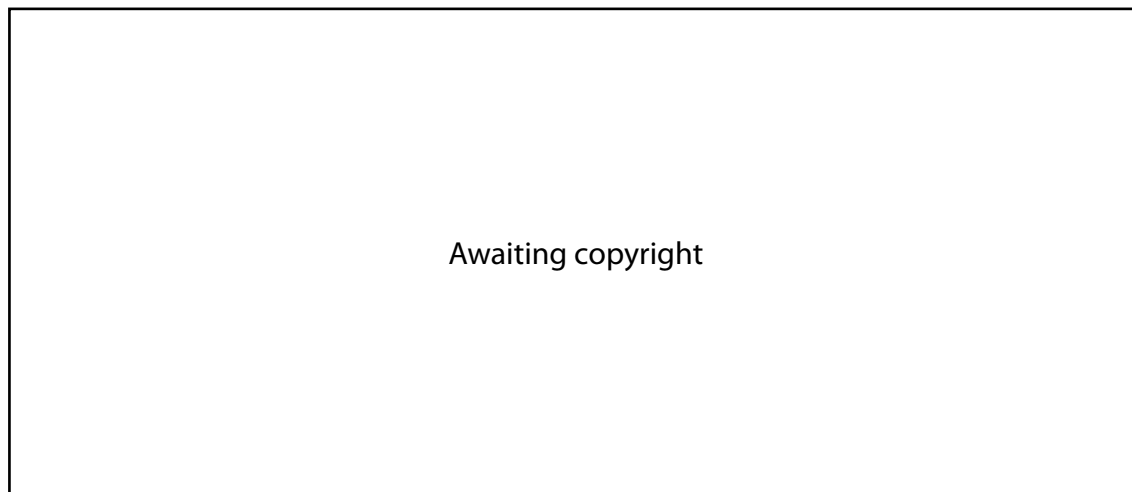
Question 24 (6 marks)

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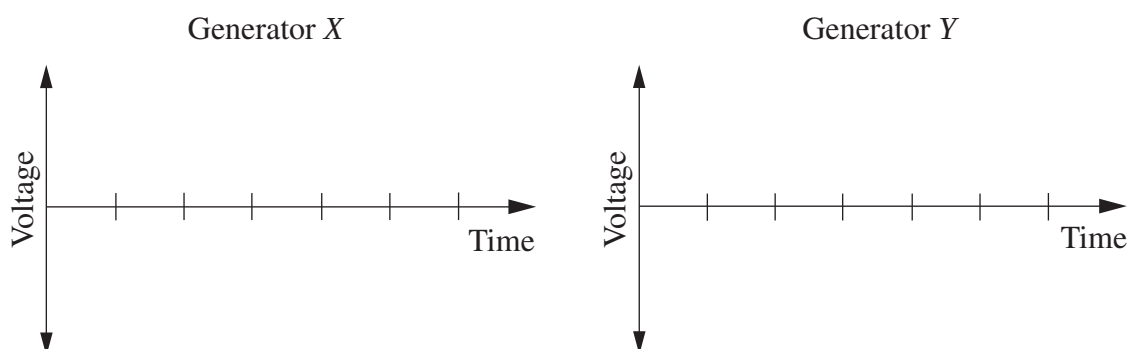
Question 25 (5 marks)

The diagrams show two different types of generator spinning at the same number of revolutions per minute. The difference between the two generators is in the way they are connected to the external circuits.



- (a) On the axes below, sketch a voltage-time graph for each generator.

2



- (b) Explain how the difference in connection to the external circuit accounts for the different output voltages.

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Centre Number

Section I (continued)

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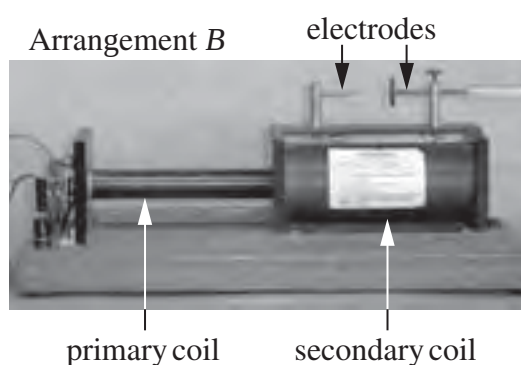
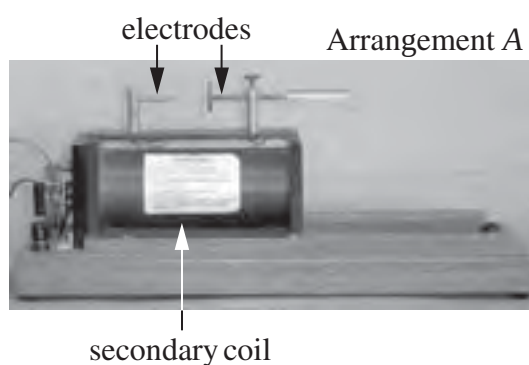
Student Number

Marks

Question 26 (3 marks)

An induction coil is a type of transformer that allows a small voltage to be stepped up to a higher voltage. An induction coil consists of a primary coil wound around an iron core and a secondary coil. The secondary coil can be moved sideways so that different lengths of the iron core are within the secondary coil.

The photographs show an induction coil with the secondary coil in two different arrangements with the power supply turned off. At sufficiently high voltages a spark can be produced between the secondary coil electrodes.



- (a) Which arrangement would produce a spark when the power supply is turned on? 1
Justify your choice.

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- (b) Explain how different voltages are induced when the secondary coil is moved to different positions. 2

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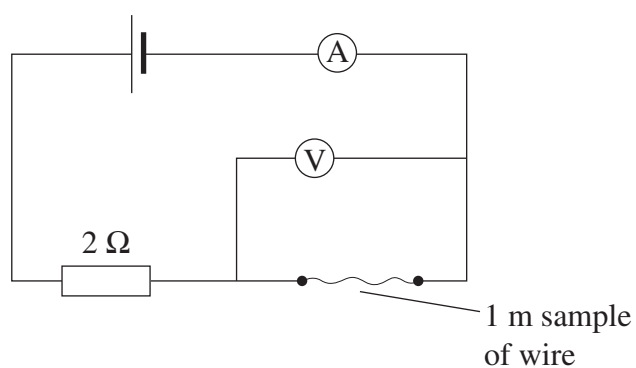
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Question 27 (6 marks)

A student was given a sample of wire *X* and a sample of wire *Y*. The wires looked identical. However, one was pure chromium and the other was nichrome, an alloy containing chromium and nickel.

To differentiate between the two wires, the student set up the circuit below and obtained the results shown in the table.

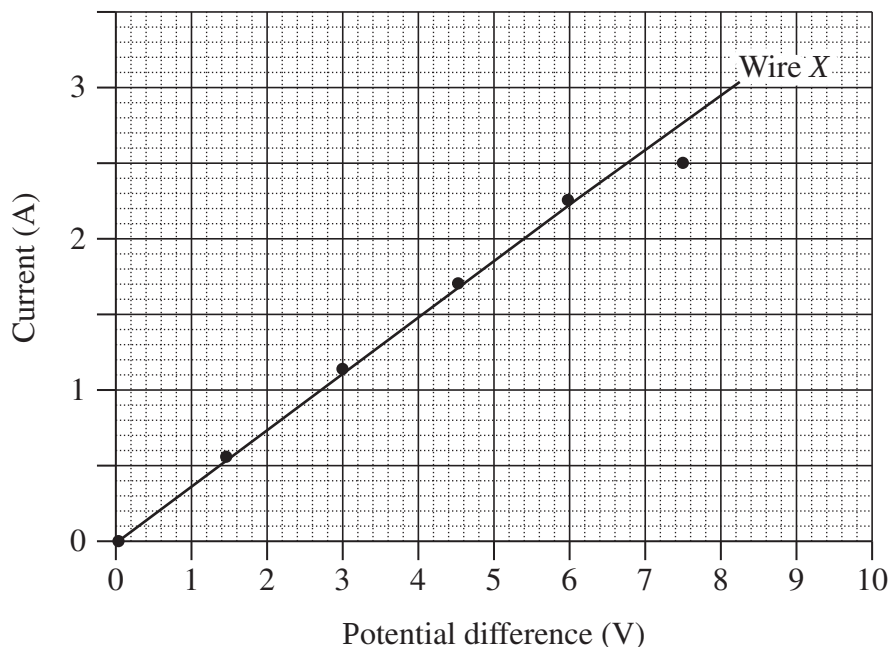


Potential difference (V)	Current (A)	
	Wire X	Wire Y
0	0	0
1.5	0.57	0.20
3.0	1.14	0.39
4.5	1.71	0.59
6.0	2.28	0.79
7.5	2.50	0.99

Question 27 continues on page 21

Question 27 (continued)

- (a) The data for wire *X* has been plotted on the graph below. Plot the data, including a trend line, for wire *Y* on the same graph. 2



- (b) Calculate the resistance of wire *Y*. 1

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- (c) Which sample of wire was pure chromium? Justify your response with reference to your graph. 2

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- (d) When the data for wire *X* was plotted, one data point was considered inconsistent and was disregarded when drawing the trend line for calculating its resistance. 1

Suggest a physical reason why this data point is inconsistent with the trend line.

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End of Question 27

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Physics

Section II

25 marks

Attempt ONE question from Questions 28–32

Allow about 45 minutes for this section

Answer the question in a writing booklet. Extra writing booklets are available.

Show all relevant working in questions involving calculations.

	Pages
Question 28 Geophysics	24–26
Question 29 Medical Physics	27
Question 30 Astrophysics	28–29
Question 31 From Quanta to Quarks	30–31
Question 32 The Age of Silicon	32–34

Question 28 — Geophysics (25 marks)

- (a) The table lists some of the principal methods used in geophysics, a property on which each method is based and an application of each method.

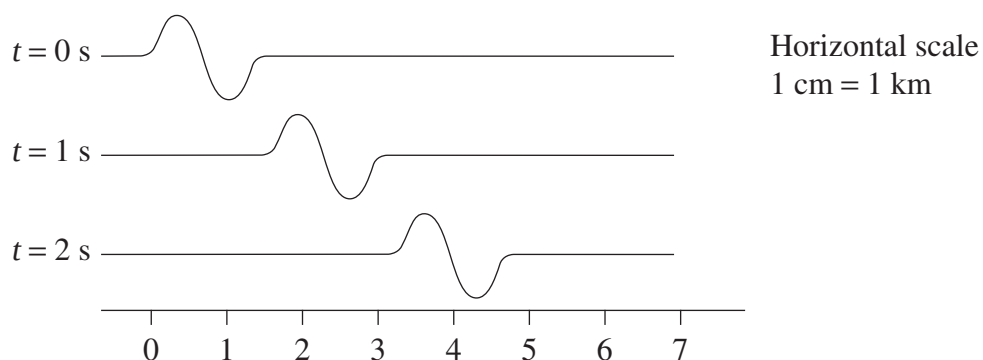
<i>Method used in geophysics</i>	<i>Property of earth materials</i>	<i>Application</i>
Magnetic	Magnetism	Plate tectonics
Gravitational	Density	<i>X</i>
Electrical	<i>Y</i>	Water location
Seismic	Elasticity of medium	<i>Z</i>

- (i) From the table, what do the letters *X*, *Y* and *Z* represent? **3**
- (ii) For any one of the principal methods used in geophysics describe how the type of information generated can be used to advance our understanding of Earth. **3**

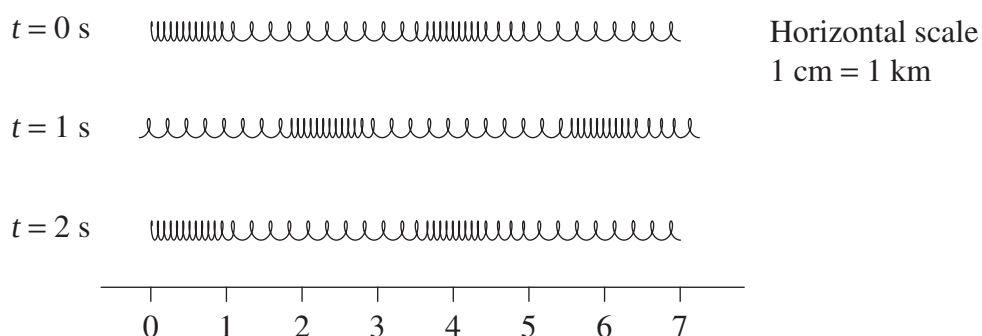
Question 28 continues on page 25

Question 28 (continued)

- (b) An *S* wave can be modelled by a transverse pulse sent along a string as indicated below.



A *P* wave can be modelled by a compression wave sent along a slinky spring as indicated below.



- (i) Calculate the speeds of the *S* wave and the *P* wave shown. 3
- (ii) Explain how *S* waves and *P* waves are reflected and refracted at an interface. 4

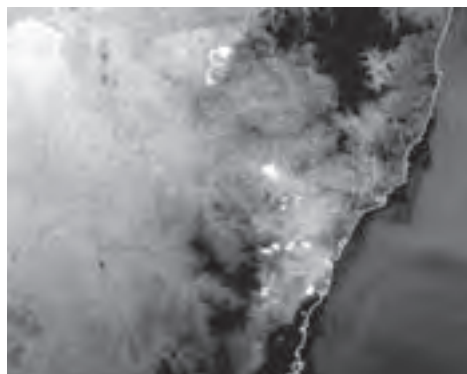
Question 28 continues on page 26

Question 28 (continued)

- (c) The CSIRO Remote Sensing Project used images from the NOAA satellites to produce the following scenes of the NSW bushfires in December 1997. The two images were taken simultaneously using different techniques.



visible image



thermal image

Reproduced with the permission of CSIRO

- (i) With reference to the two images of the scene, explain the underlying physical principles that result in the different images. 3
- (ii) Describe the role of remote sensing techniques in monitoring climate, pollution and natural hazards. 3
- (d) Both geophones and seismometers detect seismic activity. 6

Compare the structure and function of these devices and the information they provide about the large-scale structure of the Earth.

End of Question 28

Question 29 — Medical Physics (25 marks)

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|------|--|----------|
| (a) | (i) Account for the production and detection of ultrasound waves by the transducer of an ultrasound machine. | 3 |
| | (ii) Explain what happens to ultrasound waves as they travel through body tissues and return to the transducer. | 3 |
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| (b) | (i) Outline TWO uses of endoscopy. | 2 |
| | (ii) Using diagrams, distinguish between the coherent and incoherent bundles of optical fibres and their roles in endoscopy. | 3 |
| | (iii) Outline ONE advantage of endoscopy over alternative surgical procedures. | 1 |
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 | | |
| (c) | (i) Contrast the advantages of bone scans with the advantages of X-ray images when examining bones. | 3 |
| | (ii) Describe how X-rays are produced. | 2 |
| | (iii) Describe the properties of a radiopharmaceutical substance that make it suitable for producing a bone scan. | 2 |
|
 | | |
| (d) | Explain how different medical imaging techniques use tomography to improve our diagnostic abilities. | 6 |

End of Question 29

Question 30 — Astrophysics (25 marks)

- (a) The analysis of electromagnetic radiation is widely used by astronomers.
- (i) Contrast emission and absorption spectra in terms of how they are produced. 3
 - (ii) Describe the physical characteristics of stars and their motion that can be revealed by spectroscopy. 3

- (b) The table shows some photometric measurements of certain stars.

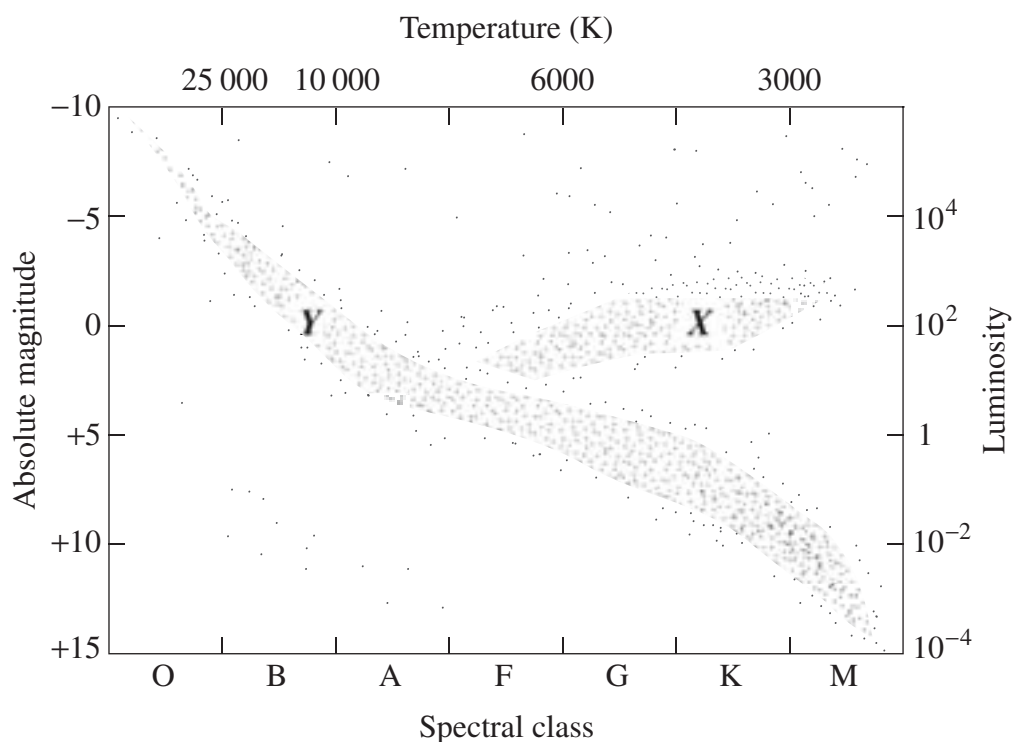
<i>Star</i>	<i>Apparent magnitude</i>	<i>Absolute magnitude</i>	<i>Colour index</i>
Bellatrix	+1.64	−2.72	−0.22
Sirius A	−1.47	+1.42	+0.01
Regulus A	+1.35	−0.52	−0.11
Betelgeuse	+0.58	−5.14	+1.85

- (i) How much brighter is Sirius A than Bellatrix when viewed from Earth? 2
- (ii) Calculate the distance from Earth to Regulus A. 2
- (iii) Explain why cooler stars have a more positive colour index than hotter stars. 3

Question 30 continues on page 29

Question 30 (continued)

- (c) (i) Describe the physical processes that precede nuclear fusion reactions in a newly formed star. 2
- (ii) Compare the nuclear reactions occurring in stars located at positions *X* and *Y* on the HR diagram below. 2

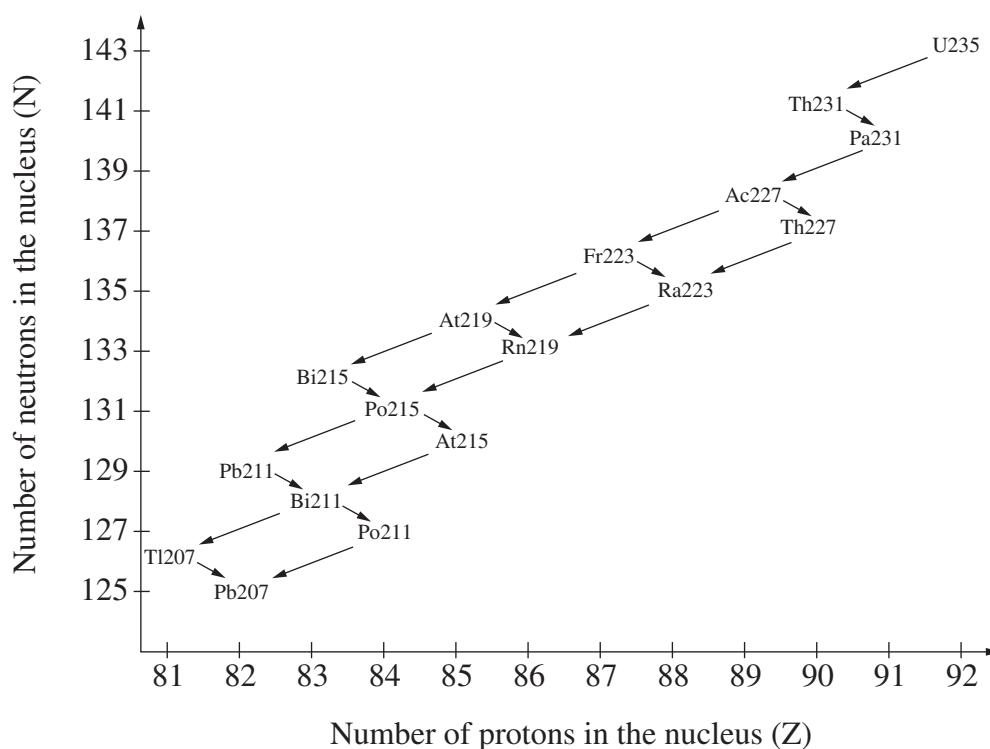


- (iii) Draw a flowchart summarising the possible pathways a red giant could follow as it evolves. 2
- (d) Explain how observations of binary and variable stars can be used to infer physical properties of these stars. 6

End of Question 30

Question 31 — From Quanta to Quarks (25 marks)

- (a) (i) Outline how you would conduct a first-hand investigation to observe the visible components of the hydrogen emission spectrum. 2
- (ii) How would the results from this investigation support Bohr's model of the atom? 2
- (iii) Outline ONE feature of atomic emission spectra that cannot be explained by Bohr's model. 2
- (b) Nuclear transmutations caused by natural radioactivity can be represented in diagrams such as the one shown. Each symbol represents a radioactive element and each arrow represents a transmutation.



- (i) How many protons and how many neutrons are there in the nucleus of a Thorium-227 atom? 1
- (ii) Write the equation for the α -decay of Francium-223. 2

Question 31 continues on page 31

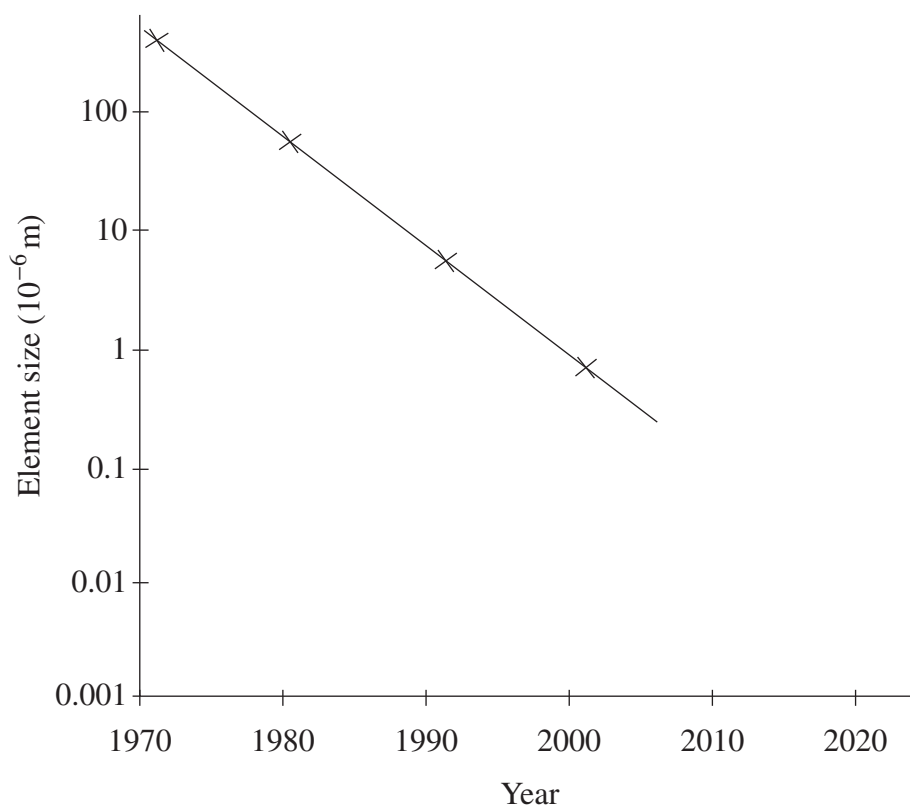
Question 31 (continued)

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|-----|------|---|----------|
| (c) | (i) | An atom of Carbon-12 has 6 protons and 6 neutrons in its nucleus. The mass of a Carbon-12 atom is 12.000 atomic mass unit. Show that the mass defect of one Carbon-12 atom is 0.097 atomic mass unit. | 3 |
| | (ii) | How much energy is this mass defect equivalent to? | 1 |
| (d) | (i) | Use a diagram to outline one way in which physicists obtain particles with the appropriate energy to investigate the structure of matter. | 2 |
| | (ii) | Describe the key features and components of the standard model of matter. | 4 |
| (e) | | Use the work of TWO physicists to explain how the combination of ideas led to new directions in scientific thinking about atomic structure. | 6 |

End of Question 31

Question 32 — The Age of Silicon (25 marks)

- (a) The graph below shows how the size of integrated circuit elements has changed over the interval 1970–2000.

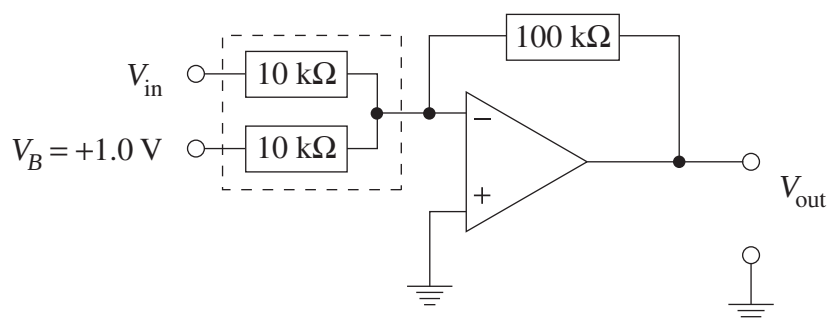


- (i) Explain the effect that this trend has had on computer performance. **3**
- (ii) Comment on the validity of using this data to predict integrated circuit element size in 2040. **2**

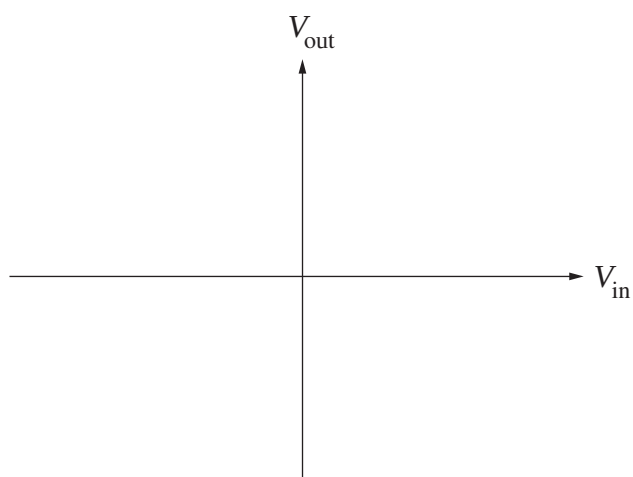
Question 32 continues on page 33

Question 32 (continued)

- (b) An ideal differential-input operational amplifier is connected into the following circuit.



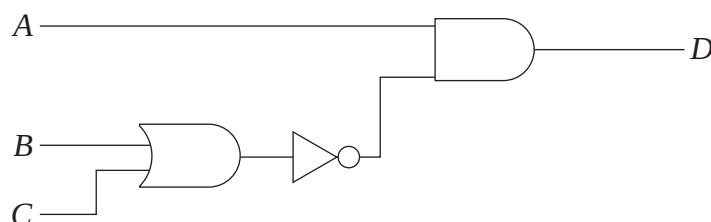
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|-------|---|---|
| (i) | Describe the properties of an ideal operational amplifier. | 2 |
| (ii) | Identify the function of the 100 kΩ resistor in this circuit. | 1 |
| (iii) | Identify the function of the portion of the circuit enclosed in the dashed box. | 1 |
| (iv) | Copy the axes below into your writing booklet and sketch the V_{out} vs V_{in} transfer characteristic of this amplifier. | 3 |



Question 32 continues on page 34

Question 32 (continued)

- (c) In recent years, torches using LEDs rather than incandescent bulbs have become commonly available.
- (i) Describe the internal structure and operation of a typical LED. 2
- (ii) Explain why LEDs are preferable to incandescent bulbs in this application. 2
- (d) For the logic circuit below, construct a truth table showing the output D for all possible combinations of inputs on A , B and C . 3



- (e) Advances in computer technology based on high-speed digital integrated circuits have had a huge impact on the design of electronics. However, analogue transducers still play an important role in many modern circuits. 6

Explain these statements, providing examples from modern electronics.

End of paper

DATA SHEET

Charge on electron, q_e	$-1.602 \times 10^{-19} \text{ C}$
Mass of electron, m_e	$9.109 \times 10^{-31} \text{ kg}$
Mass of neutron, m_n	$1.675 \times 10^{-27} \text{ kg}$
Mass of proton, m_p	$1.673 \times 10^{-27} \text{ kg}$
Speed of sound in air	340 m s^{-1}
Earth's gravitational acceleration, g	9.8 m s^{-2}
Speed of light, c	$3.00 \times 10^8 \text{ m s}^{-1}$
Magnetic force constant, $\left(k \equiv \frac{\mu_0}{2\pi} \right)$	$2.0 \times 10^{-7} \text{ N A}^{-2}$
Universal gravitational constant, G	$6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Mass of Earth	$6.0 \times 10^{24} \text{ kg}$
Planck constant, h	$6.626 \times 10^{-34} \text{ J s}$
Rydberg constant, R (hydrogen)	$1.097 \times 10^7 \text{ m}^{-1}$
Atomic mass unit, u	$1.661 \times 10^{-27} \text{ kg}$ $931.5 \text{ MeV}/c^2$
1 eV	$1.602 \times 10^{-19} \text{ J}$
Density of water, ρ	$1.00 \times 10^3 \text{ kg m}^{-3}$
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

FORMULAE SHEET

$$v = f\lambda$$

$$I \propto \frac{1}{d^2}$$

$$\frac{v_1}{v_2} = \frac{\sin i}{\sin r}$$

$$E = \frac{F}{q}$$

$$R = \frac{V}{I}$$

$$P = VI$$

$$\text{Energy} = VIt$$

$$v_{\text{av}} = \frac{\Delta r}{\Delta t}$$

$$a_{\text{av}} = \frac{\Delta v}{\Delta t} \text{ therefore } a_{\text{av}} = \frac{v - u}{t}$$

$$\Sigma F = ma$$

$$F = \frac{mv^2}{r}$$

$$E_k = \frac{1}{2}mv^2$$

$$W = Fs$$

$$p = mv$$

$$\text{Impulse} = Ft$$

$$E_p = -G \frac{m_1 m_2}{r}$$

$$F = mg$$

$$v_x^2 = u_x^2$$

$$v = u + at$$

$$v_y^2 = u_y^2 + 2a_y \Delta y$$

$$\Delta x = u_x t$$

$$\Delta y = u_y t + \frac{1}{2}a_y t^2$$

$$\frac{r^3}{T^2} = \frac{GM}{4\pi^2}$$

$$F = \frac{Gm_1 m_2}{d^2}$$

$$E = mc^2$$

$$l_v = l_0 \sqrt{1 - \frac{v^2}{c^2}}$$

$$t_v = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$m_v = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

FORMULAE SHEET

$$\frac{F}{l} = k \frac{I_1 I_2}{d}$$

$$d = \frac{1}{p}$$

$$F = BIl \sin \theta$$

$$M = m - 5 \log \left(\frac{d}{10} \right)$$

$$\tau = Fd$$

$$\frac{I_A}{I_B} = 100^{(m_B - m_A)/5}$$

$$\tau = nBIA \cos \theta$$

$$m_1 + m_2 = \frac{4\pi^2 r^3}{GT^2}$$

$$\frac{V_p}{V_s} = \frac{n_p}{n_s}$$

$$F = qvB \sin \theta$$

$$\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

$$E = \frac{V}{d}$$

$$\lambda = \frac{h}{mv}$$

$$E = hf$$

$$c = f\lambda$$

$$A_0 = \frac{V_{\text{out}}}{V_{\text{in}}}$$

$$Z = \rho v$$

$$\frac{V_{\text{out}}}{V_{\text{in}}} = - \frac{R_f}{R_i}$$

$$\frac{I_r}{I_0} = \frac{[Z_2 - Z_1]^2}{[Z_2 + Z_1]^2}$$

PERIODIC TABLE OF THE ELEMENTS

1 H 1.008 Hydrogen		KEY										2 He 4.003 Helium																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
3 Li 6.941 Lithium		4 Be 9.012 Beryllium		Atomic Number		79 Au 197.0 Gold		Symbol of element		9 F 19.00 Fluorine		8 O 16.00 Oxygen		7 N 14.01 Nitrogen		6 C 12.01 Carbon		5 B 10.81 Boron																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
		12 Mg 24.31 Magnesium		Atomic Weight		Gold		Name of element		Fluorine		Oxygen		Nitrogen		Carbon		Boron																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
11 Na 22.99 Sodium		18 Ar 39.95 Argon		25 Mn 54.94 Manganese		26 Fe 55.85 Iron		27 Co 58.93 Cobalt		28 Ni 58.69 Nickel		29 Cu 63.55 Copper		30 Zn 65.41 Zinc		31 Ga 69.72 Gallium		32 Ge 72.64 Germanium		33 As 74.92 Arsenic		34 Se 78.96 Selenium		35 Br 79.90 Bromine		36 Kr 83.80 Krypton		37 Rb 85.47 Rubidium		38 Sr 87.62 Strontium		39 Y 88.91 Yttrium		40 Zr 91.22 Zirconium		41 Nb 92.91 Niobium		42 Mo 95.94 Molybdenum		43 Tc [97.91] Technetium		44 Ru 101.1 Ruthenium		45 Rh 102.9 Rhodium		46 Pd 106.4 Palladium		47 Ag 107.9 Silver		48 Cd 112.4 Cadmium		49 In 114.8 Indium		50 Sn 118.7 Tin		51 Sb 121.8 Antimony		52 Te 127.6 Tellurium		53 I 126.9 Iodine		54 Xe 131.3 Xenon		55 Cs 132.9 Caesium		56 Ba 137.3 Barium		57-71 Lanthanoids		72 Hf 178.5 Hafnium		73 Ta 180.9 Tantalum		74 W 183.8 Tungsten		75 Re 186.2 Rhenium		76 Os 190.2 Osmium		77 Ir 192.2 Iridium		78 Pt 195.1 Platinum		79 Au 197.0 Gold		80 Hg 200.6 Mercury		81 Tl 204.4 Thallium		82 Pb 207.2 Lead		83 Bi 209.0 Bismuth		84 Po [209.0] Polonium		85 At [210.0] Astatine		86 Rn [222.0] Radon		87 Fr [223] Francium		88 Ra [226] Radium		89-103 Actinoids		104 Rf [261] Rutherfordium		105 Db [262] Dubnium		106 Sg [266] Seaborgium		107 Bh [264] Bohrium		108 Hs [277] Hassium		109 Mt [268] Meitnerium		110 Ds [271] Darmstadtium		111 Rg [272] Roentgenium		112 Nh [284] Nihonium		113 Fl [286] Flerovium		114 Mc [289] Moscovium		115 Lv [293] Livermorium		116 Ts [294] Tennessine		117 Og [294] Oganesson		118 Uu [286] Ununennium		119 Uub [287] Unbibium		120 Uut [288] Untrium		121 Uuq [289] Unquadrium		122 Uubk [290] Unbikium		123 Uufl [291] Unbiflermium		124 Uuhc [292] Unbihexium		125 Uuhk [293] Unbihexium		126 Uuhl [294] Unbihexium		127 Uuhv [295] Unbihexium		128 Uuht [296] Unbihexium		129 Uuhv [297] Unbihexium		130 Uuht [298] Unbihexium		131 Uuhv [299] Unbihexium		132 Uuht [300] Unbihexium		133 Uuhv [301] Unbihexium		134 Uuht [302] Unbihexium		135 Uuhv [303] Unbihexium		136 Uuht [304] Unbihexium		137 Uuhv [305] Unbihexium		138 Uuht [306] Unbihexium		139 Uuhv [307] Unbihexium		140 Uuht [308] Unbihexium		141 Uuhv [309] Unbihexium		142 Uuht [310] Unbihexium		143 Uuhv [311] Unbihexium		144 Uuht [312] Unbihexium		145 Uuhv [313] Unbihexium		146 Uuht [314] Unbihexium		147 Uuhv [315] Unbihexium		148 Uuht [316] Unbihexium		149 Uuhv [317] Unbihexium		150 Uuht [318] Unbihexium		151 Uuhv [319] Unbihexium		152 Uuht [320] Unbihexium		153 Uuhv [321] Unbihexium		154 Uuht [322] Unbihexium		155 Uuhv [323] Unbihexium		156 Uuht [324] Unbihexium		157 Uuhv [325] Unbihexium		158 Uuht [326] Unbihexium		159 Uuhv [327] Unbihexium		160 Uuht [328] Unbihexium		161 Uuhv [329] Unbihexium		162 Uuht [330] Unbihexium		163 Uuhv [331] Unbihexium		164 Uuht [332] Unbihexium		165 Uuhv [333] Unbihexium		166 Uuht [334] Unbihexium		167 Uuhv [335] Unbihexium		168 Uuht [336] Unbihexium		169 Uuhv [337] Unbihexium		170 Uuht [338] Unbihexium		171 Uuhv [339] Unbihexium		172 Uuht [340] Unbihexium		173 Uuhv [341] Unbihexium		174 Uuht [342] Unbihexium		175 Uuhv [343] Unbihexium		176 Uuht [344] Unbihexium		177 Uuhv [345] Unbihexium		178 Uuht [346] Unbihexium		179 Uuhv [347] Unbihexium		180 Uuht [348] Unbihexium		181 Uuhv [349] Unbihexium		182 Uuht [350] Unbihexium		183 Uuhv [351] Unbihexium		184 Uuht [352] Unbihexium		185 Uuhv [353] Unbihexium		186 Uuht [354] Unbihexium		187 Uuhv [355] Unbihexium		188 Uuht [356] Unbihexium		189 Uuhv [357] Unbihexium		190 Uuht [358] Unbihexium		191 Uuhv [359] Unbihexium		192 Uuht [360] Unbihexium		193 Uuhv [361] Unbihexium		194 Uuht [362] Unbihexium		195 Uuhv [363] Unbihexium		196 Uuht [364] Unbihexium		197 Uuhv [365] Unbihexium		198 Uuht [366] Unbihexium		199 Uuhv [367] Unbihexium		200 Uuht [368] Unbihexium		201 Uuhv [369] Unbihexium		202 Uuht [370] Unbihexium		203 Uuhv [371] Unbihexium		204 Uuht [372] Unbihexium		205 Uuhv [373] Unbihexium		206 Uuht [374] Unbihexium		207 Uuhv [375] Unbihexium		208 Uuht [376] Unbihexium		209 Uuhv [377] Unbihexium		210 Uuht [378] Unbihexium		211 Uuhv [379] Unbihexium		212 Uuht [380] Unbihexium		213 Uuhv [381] Unbihexium		214 Uuht [382] Unbihexium		215 Uuhv [383] Unbihexium		216 Uuht [384] Unbihexium		217 Uuhv [385] Unbihexium		218 Uuht [386] Unbihexium		219 Uuhv [387] Unbihexium		220 Uuht [388] Unbihexium		221 Uuhv [389] Unbihexium		222 Uuht [390] Unbihexium		223 Uuhv [391] Unbihexium		224 Uuht [392] Unbihexium		225 Uuhv [393] Unbihexium		226 Uuht [394] Unbihexium		227 Uuhv [395] Unbihexium		228 Uuht [396] Unbihexium		229 Uuhv [397] Unbihexium		230 Uuht [398] Unbihexium		231 Uuhv [399] Unbihexium		232 Uuht [400] Unbihexium		233 Uuhv [401] Unbihexium		234 Uuht [402] Unbihexium		235 Uuhv [403] Unbihexium		236 Uuht [404] Unbihexium		237 Uuhv [405] Unbihexium		238 Uuht [406] Unbihexium		239 Uuhv [407] Unbihexium		240 Uuht [408] Unbihexium		241 Uuhv [409] Unbihexium		242 Uuht [410] Unbihexium		243 Uuhv [411] Unbihexium		244 Uuht [412] Unbihexium		245 Uuhv [413] Unbihexium		246 Uuht [414] Unbihexium		247 Uuhv [415] Unbihexium		248 Uuht [416] Unbihexium		249 Uuhv [417] Unbihexium		250 Uuht [418] Unbihexium		251 Uuhv [419] Unbihexium		252 Uuht [420] Unbihexium		253 Uuhv [421] Unbihexium		254 Uuht [422] Unbihexium		255 Uuhv [423] Unbihexium		256 Uuht [424] Unbihexium		257 Uuhv [425] Unbihexium		258 Uuht [426] Unbihexium		259 Uuhv [427] Unbihexium		260 Uuht [428] Unbihexium		261 Uuhv [429] Unbihexium		262 Uuht [430] Unbihexium		263 Uuhv [431] Unbihexium		264 Uuht [432] Unbihexium		265 Uuhv [433] Unbihexium		266 Uuht [434] Unbihexium		267 Uuhv [435] Unbihexium		268 Uuht [436] Unbihexium		269 Uuhv [437] Unbihexium		270 Uuht [438] Unbihexium		271 Uuhv [439] Unbihexium		272 Uuht [440] Unbihexium		273 Uuhv [441] Unbihexium		274 Uuht [442] Unbihexium		275 Uuhv [443] Unbihexium		276 Uuht [444] Unbihexium		277 Uuhv [445] Unbihexium		278 Uuht [446] Unbihexium		279 Uuhv [447] Unbihexium		280 Uuht [448] Unbihexium		281 Uuhv [449] Unbihexium		282 Uuht [450] Unbihexium		283 Uuhv [451] Unbihexium		284 Uuht [452] Unbihexium		285 Uuhv [453] Unbihexium		286 Uuht [454] Unbihexium		287 Uuhv [455] Unbihexium		288 Uuht [456] Unbihexium		289 Uuhv [457] Unbihexium		290 Uuht [458] Unbihexium		291 Uuhv [459] Unbihexium		292 Uuht [460] Unbihexium		293 Uuhv [461] Unbihexium		294 Uuht [462] Unbihexium		295 Uuhv [463] Unbihexium		296 Uuht [464] Unbihexium		297 Uuhv [465] Unbihexium		298 Uuht [466] Unbihexium		299 Uuhv [467] Unbihexium		300 Uuht [468] Unbihexium		301 Uuhv [469] Unbihexium		302 Uuht [470] Unbihexium		303 Uuhv [471] Unbihexium		304 Uuht [472] Unbihexium		305 Uuhv [473] Unbihexium		306 Uuht [474] Unbihexium		307 Uuhv [475] Unbihexium		308 Uuht [476] Unbihexium		309 Uuhv [477] Unbihexium		310 Uuht [478] Unbihexium		311 Uuhv [479] Unbihexium		312 Uuht [480] Unbihexium		313 Uuhv [481] Unbihexium		314 Uuht [482] Unbihexium		315 Uuhv [483] Unbihexium		316 Uuht [484] Unbihexium		317 Uuhv [485] Unbihexium		318 Uuht [486] Unbihexium		319 Uuhv [487] Unbihexium		320 Uuht [488] Unbihexium		321 Uuhv [489] Unbihexium		322 Uuht [490] Unbihexium		323 Uuhv [491] Unbihexium		324 Uuht [492] Unbihexium		325 Uuhv [493] Unbihexium		326 Uuht [494] Unbihexium		327 Uuhv [495] Unbihexium		328 Uuht [496] Unbihexium		329 Uuhv [497] Unbihexium		330 Uuht [498] Unbihexium		331 Uuhv [499] Unbihexium		332 Uuht [500] Unbihexium		333 Uuhv [501] Unbihexium		334 Uuht [502] Unbihexium		335 Uuhv [503] Unbihexium		336 Uuht [504] Unbihexium		337 Uuhv [505] Unbihexium		338 Uuht [506] Unbihexium		339 Uuhv [507] Unbihexium		340 Uuht [508] Unbihexium		341 Uuhv [509] Unbihexium		342 Uuht [510] Unbihexium		343 Uuhv [511] Unbihexium		344 Uuht [512] Unbihexium		345 Uuhv [513] Unbihexium		346 Uuht [514] Unbihexium		347 Uuhv [515] Unbihexium		348 Uuht [516] Unbihexium		349 Uuhv [517] Unbihexium		350 Uuht [518] Unbihexium		351 Uuhv [519] Unbihexium		352 Uuht [520] Unbihexium		353 Uuhv [521] Unbihexium		354 Uuht [522] Unbihexium		355 Uuhv [523] Unbihexium		356 Uuht [524] Unbihexium		357 Uuhv [525] Unbihexium		358 Uuht [526] Unbihexium		359 Uuhv [527] Unbihexium		360 Uuht [528] Unbihexium		361 Uuhv [529] Unbihexium		362 Uuht [530] Unbihexium		363 Uuhv [531] Unbihexium		364 Uuht [532] Unbihexium		365 Uuhv [533] Unbihexium		366 Uuht [534] Unbihexium		367 Uuhv [535] Unbihexium		368 Uuht [536] Unbihexium		369 Uuhv [537] Unbihexium		370 Uuht [538] Unbihexium		371 Uuhv [539] Unbihexium		372 Uuht [540] Unbihexium		373 Uuhv [541] Unbihexium		374 Uuht [542] Unbihexium		375 Uuhv [543] Unbihexium		376 Uuht [544] Unbihexium		377 Uuhv [545] Unbihexium		378 Uuht [546] Unbihexium		379 Uuhv [547] Unbihexium		380 Uuht [548] Unbihexium		381 Uuhv [549] Unbihexium		382 Uuht [550] Unbihexium		383 Uuhv [551] Unbihexium		384 Uuht [552] Unbihexium		385 Uuhv [553] Unbihexium		386 Uuht [554] Unbihexium		387 Uuhv [555] Unbihexium		388 Uuht [556] Unbihexium		389 Uuhv [557] Unbihexium		390 Uuht [558] Unbihexium		391 Uuhv [559] Unbihexium		392 Uuht [560] Unbihexium		393 Uuhv [561] Unbihexium		394 Uuht [562] Unbihexium		395 Uuhv [563] Unbihexium		396 Uuht [564] Unbihexium		397 Uuhv [565] Unbihexium		398 Uuht [566] Unbihexium		399 Uuhv [567] Unbihexium		400 Uuht [568] Unbihexium		401 Uuhv [569] Unbihexium		402 Uuht [570] Unbihexium		403 Uuhv [571] Unbihexium		404 Uuht [572] Unbihexium		405 Uuhv [573] Unbihexium		406 Uuht [574] Unbihexium		407 Uuhv [575] Unbihexium		408 Uuht [576] Unbihexium		409 Uuhv [577] Unbihexium		410 Uuht [578] Unbihexium		411 Uuhv [579] Unbihexium		412 Uuht [580] Unbihexium		413 Uuhv [581] Unbihexium		414 Uuht [582] Unbihexium		415 Uuhv [583] Unbihexium		416 Uuht [584] Unbihexium		417 Uuhv [585] Unbihexium		418 Uuht [586] Unbihexium		419 Uuhv [587] Unbihexium		420 Uuht [588] Unbihexium		421 Uuhv [589] Unbihexium		422 Uuht [590] Unbihexium		423 Uuhv [591] Unbihexium		424 Uuht [592] Unbihexium		425 Uuhv [593] Unbihexium		426 Uuht [594] Unbihexium		427 Uuhv [595] Unbihexium			